A POWERBI PROJECT

ON

CUSTOMER SALES REPORT

Submitted in partial fulfilment of the requirement for the

BACHELOR OF TECHNOLOGY

IN

INFORMATION TECHNOLOGY

BY

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Life. Transformed.

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1. INTRODUCTION

The Customer Sales Report is a data analysis and visualization project developed using Microsoft Power BI. The project analyzes customer shopping transactions based on categories, revenue, quantity, gender, payment methods, shopping malls, and transaction dates.

The main purpose of this project is to transform raw customer transaction data into an interactive dashboard that helps users understand sales performance and identify important business trends.

Power BI was used for data cleaning, transformation, data modeling, analysis, and visualization. The final dashboard provides important Key Performance Indicators (KPIs) and charts for analyzing customer sales data.

2. PROBLEM STATEMENT

The raw customer shopping dataset contains a large number of transaction records.
Analyzing such data manually can be difficult and time-consuming.
The major problems are:

- Large volume of transaction records makes manual analysis difficult.
- Raw data is not directly suitable for visualization.
- It is difficult to quickly identify high-performing product categories.
- Comparing revenue across shopping malls requires additional analysis.
- Understanding customer purchasing behavior from raw records is difficult.
- Different payment methods and customer segments need to be analyzed.
- Management needs a simple dashboard to monitor important sales KPIs.

To analyze customer shopping transaction data and develop an interactive Power BI dashboard that provides meaningful insights into revenue, customers, orders, quantity, product categories, shopping malls, gender, payment methods, and monthly sales trends.

3. AIM OF THE PROJECT

The aim of this project is to develop an interactive Power BI dashboard for analyzing customer sales data and converting raw transaction data into meaningful business insights.
The dashboard helps users:

- Monitor total revenue.
- Analyze customer and order information.
- Compare sales across product categories.
- Analyze revenue by shopping mall.
- Understand customer spending based on gender.
- Analyze preferred payment methods.
- Identify sales trends over time.
- Make data-driven business decisions.

4. OBJECTIVES

The major objectives of the project are:

1. To collect and understand the customer shopping dataset.
2. To clean and prepare the raw dataset for analysis.
3. To perform data transformation using Power Query.
4. To create calculated columns and measures where required.
5. To calculate important sales KPIs.
6. To analyze revenue by product category.
7. To analyze revenue by shopping mall.
8. To analyze revenue by gender.
9. To analyze revenue by payment method.
10. To study monthly/yearly revenue trends.
11. To create interactive charts and visualizations.

12. To develop an easy-to-understand Power BI dashboard.
13. To derive meaningful insights from the analyzed data.

5. DATASET

5.1 Dataset Name:

Customer Shopping Data

5.2 Dataset Description:

Description		Value
Total Rows	-	99,457
Total Columns	-	10
Categories	-	8
Payment Methods	-	3
Shopping Malls	-	10
Genders	-	2
Date Range	-	01-Jan-2021 to 08-Mar-2023

6. DATA SOURCE LINK

Data Source: Customer Shopping Dataset

Source Link:

https://www.kaggle.com/datasets/mehmettahiraslan/customer-shopping-dataset

7. EXPLANATION OF DATA

Column	Description	Data Type
Invoice No	Unique transaction/invoice number	Text
CustomerID	Unique customer identification number	Text
Gender	Gender of the customer	Categorical
Age	Age of the customer	Numeric
Category	Product category purchased	Categorical
Quantity	Number of products purchased	Numeric
Price	Price of the product/transaction	Numeric
Payment Method	Method used for payment	Categorical
Invoice Date	Date of the transaction	Date
Shopping Mall	Shopping mall where purchase was made	Categorical

8. ROWS AND COLUMNS

Rows:

The dataset contains:

99,457 rows

Each row represents one customer transaction.

Columns:

The dataset contains:

10 columns

These columns contain customer, transaction, product, payment, date, and shopping mall information.

9. CATEGORICAL VARIABLES AND NUMBER OF UNIQUE VALUES

Column	Number of Unique Values
Gender	2
Category	8
Payment Method	3
Shopping Mall	10
Invoice Date	797
Customer ID	99,457
Invoice No	99,457

Categories:

The dataset contains:

- Clothing
- Shoes
- Technology
- Cosmetics
- Toys
- Food & Beverage
- Books
- Souvenir

Payment Methods:

- Cash
- Credit Card
- Debit Card

10. DATA CLEANING AND TRANSFORMATION

The raw dataset was imported into **Power BI using Power Query** for data preparation.
The following cleaning and transformation operations were performed or checked:

10.1 Data Type Checking

The data types of the columns were checked and assigned appropriately.

- Invoice No → Text
- Customer ID → Text
- Gender → Text
- Age → Whole Number
- Category → Text
- Quantity → Whole Number
- Price → Decimal Number
- Payment Method → Text

- Invoice Date → Date
- Shopping Mall → Text

10.2 Missing Value Check

The dataset was checked for null or blank values.

Result:

No missing/null values were found in the dataset.

10.3 Duplicate Check

Duplicate records were checked.

Result:

No duplicate rows were identified.

Invoice numbers and customer IDs were also checked for uniqueness in this dataset.

10.4 Date Transformation

The Invoice Date column was converted from text format into a proper **Date** data type.

This allows Power BI to perform:

- Year analysis
- Month analysis
- Monthly revenue trends
- Date-based filtering

10.5 Revenue Calculation

A revenue calculation was created using:

Revenue = Quantity × Price

This calculated value is used for revenue analysis and dashboard KPIs.

11. BEFORE DATASET

Before Cleaning / Transformation:

The raw dataset contained transaction information in its original form.

Before Transformation:

- Invoice Date was treated as a date field after validation.
- Revenue was not directly available as a separate calculated field.
- Data types required checking.
- Dataset required quality checks for blanks and duplicates.
- Data needed to be prepared for Power BI analysis.
- Additional calculated measures were required for dashboard KPIs.

12. AFTER DATASET

After Cleaning and Transformation:

After the data preparation process, the dataset became suitable for analysis and visualization.

After Transformation:

- Data types were standardized.
- Invoice dates were converted to proper date format.
- Data quality was checked.
- No missing values were present.
- No duplicate rows were found.
- Revenue was calculated using Quantity × Price.
- Data was loaded into Power BI for visualization.
- Measures were created for dashboard KPIs.

Important Calculated Field:

Revenue = Quantity × Price

13. ETL OPERATIONS

ETL stands for:

Extract → Transform → Load

13.1 Extract:

The customer shopping dataset was imported into Power BI.

13.2 Transform:

The data was prepared using Power Query.

Transformation activities included:

- Checking column names
- Checking data types
- Checking missing values
- Checking duplicate records
- Converting date fields
- Creating required calculated fields
- Preparing the dataset for analysis

13.3 Load:

After transformation, the cleaned data was loaded into the Power BI data model.

ETL Flow:

Raw Dataset
↓
Extract
↓
Power Query
↓
Clean & Transform
↓
Create Calculations
↓
Load into Power BI
↓
Create Visualizations
↓
Interactive Dashboard

14. ANALYSIS

The cleaned dataset was analyzed using Power BI.

The major analysis areas are:

14.1 Revenue Analysis:

Total revenue generated from the dataset is approximately:

251.51 Million

Revenue was analyzed across:

- Category
- Shopping Mall
- Gender
- Payment Method
- Time

14.2 Customer Analysis:

The dataset contains approximately:

99K customers

Customer information was analyzed using gender, age, category, and transaction information.

14.3 Order Analysis:

Total orders/transactions:

99K

14.4 Quantity Analysis:

Total quantity sold:

299K

More precisely, the dataset contains **298,712 units**.

14.5 Average Order Value:

Average Order Value is calculated as:

Total Revenue ÷ Total Orders

Approximately:

2.53K

15. MEASURES AND CALCULATED COLUMNS

Total Revenue:

Total Revenue = SUMX('Customer Shopping Data',
'Customer Shopping Data'[Quantity] *
'Customer Shopping Data'[Price])

Total Customers:

Total Customers = DISTINCTCOUNT('Customer Shopping Data'[Customer ID])

Total Orders:

Total Orders = DISTINCTCOUNT('Customer Shopping Data'[Invoice No])

Total Quantity:

Total Quantity = SUM('Customer Shopping Data'[Quantity])

Average Order Value:

Average Order Value =
DIVIDE([Total Revenue], [Total Orders])

16. DASHBOARD

Dashboard Title:

CUSTOMER SALES REPORT

The dashboard was designed to provide a summarized view of customer sales performance.
It contains KPI cards, bar charts, column charts, pie/donut charts, and treemap visualizations.

KPI Cards:

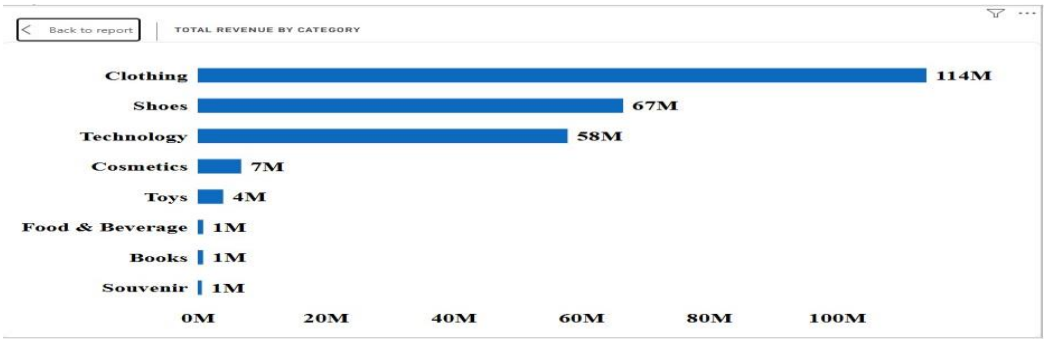
The dashboard contains five major KPIs:

1. Total Revenue – 251.51M
2. Total Customers – 99K
3. Total Orders – 99K
4. Total Quantity – 299K
5. Average Order Value – 2.53K

17. VISUALIZATIONS USED

17.1 Total Revenue by Category:

Chart Type: Bar Chart



This visualization compares the total revenue generated by each product category.

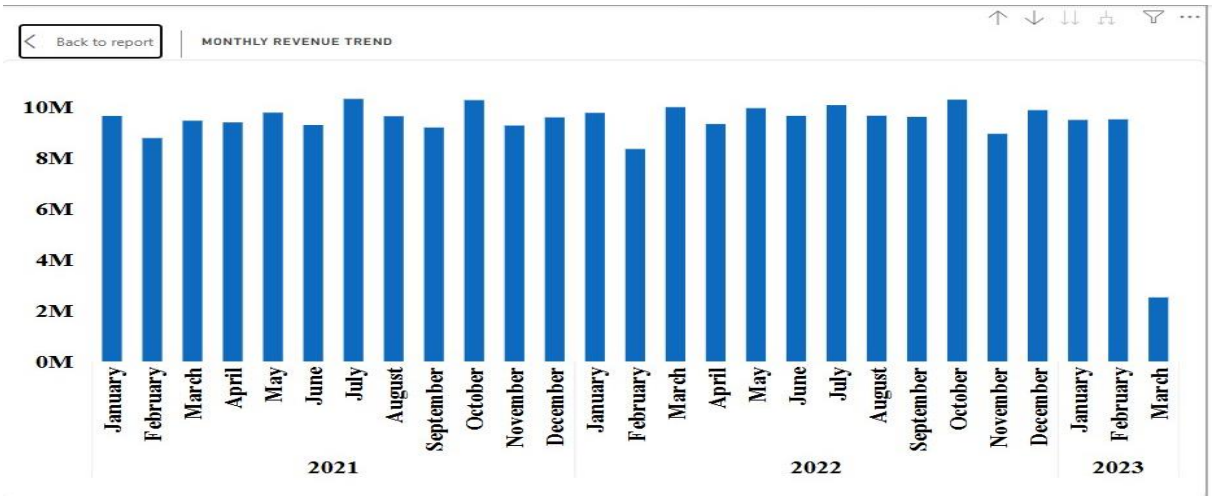
The major categories include:

- Clothing
- Shoes
- Technology
- Cosmetics
- Toys
- Food & Beverage
- Books
- Souvenir

From the dataset, Clothing generates the highest revenue, followed by Shoes and Technology.

17.2 Monthly Revenue Trend:

Chart Type: Column Chart



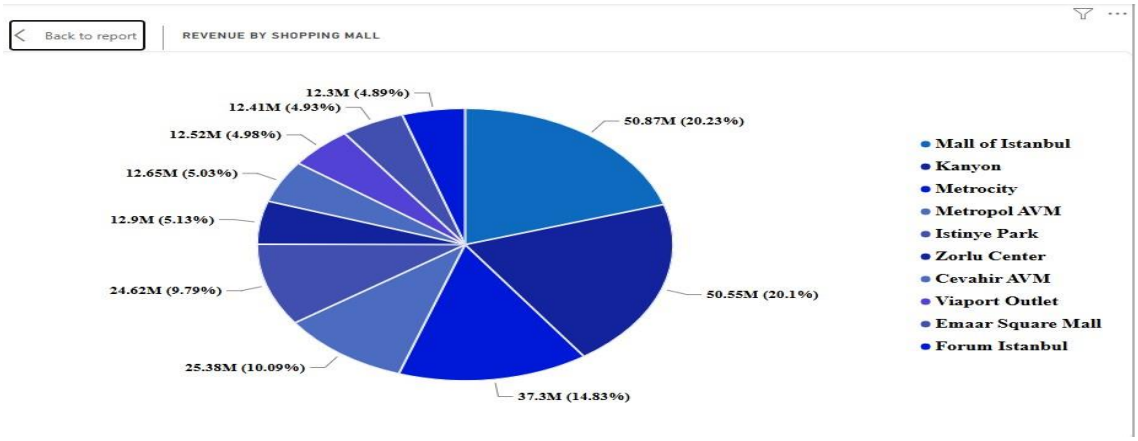
This visual is used to analyze revenue across different periods.

It helps identify:

- Revenue changes over time
- Yearly/monthly performance
- Periods with relatively higher or lower sales

17.3 REVENUE BY SHOPPING MALL:

Chart Type: Pie Chart



This chart compares revenue generated by different shopping malls.

The dataset contains 10 shopping malls.

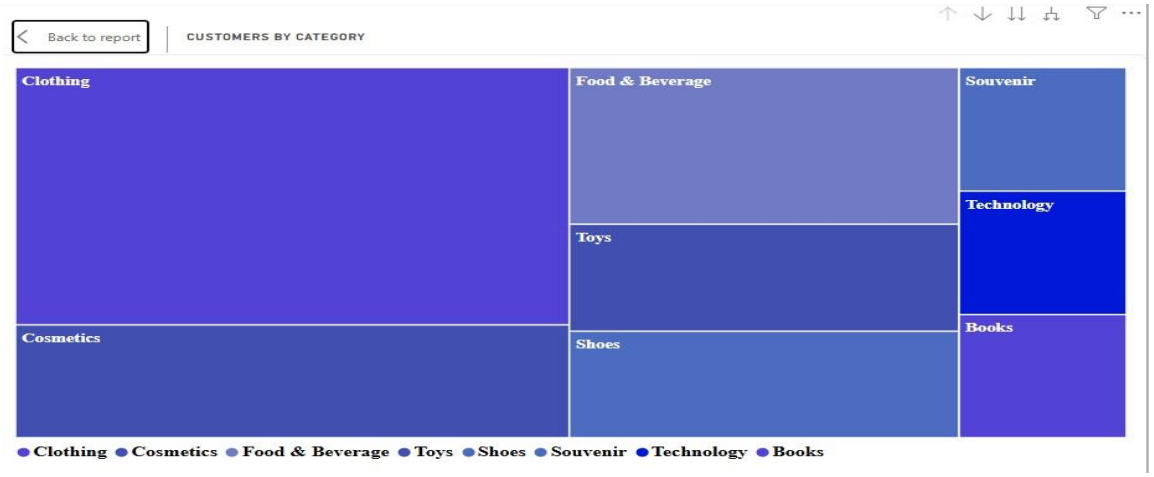
Based on the calculated revenue:

Shopping Mall	Revenue
Mall of Istanbul	50.87M
Kanyon	50.55M
Metrocity	37.30M
Metropol AVM	25.38M
Istinye Park	24.62M
Zorlu Center	12.90M
Cevahir AVM	12.65M
Viaport Outlet	12.52M
Emaar Square Mall	12.41M
Forum Istanbul	12.30M

This visualization helps compare the contribution of different shopping malls to total revenue.

17.4 CUSTOMERS BY CATEGORY:

Chart Type: Treemap



The treemap displays the distribution of customer transactions across different product categories.

It helps users quickly understand the relative size of different categories.

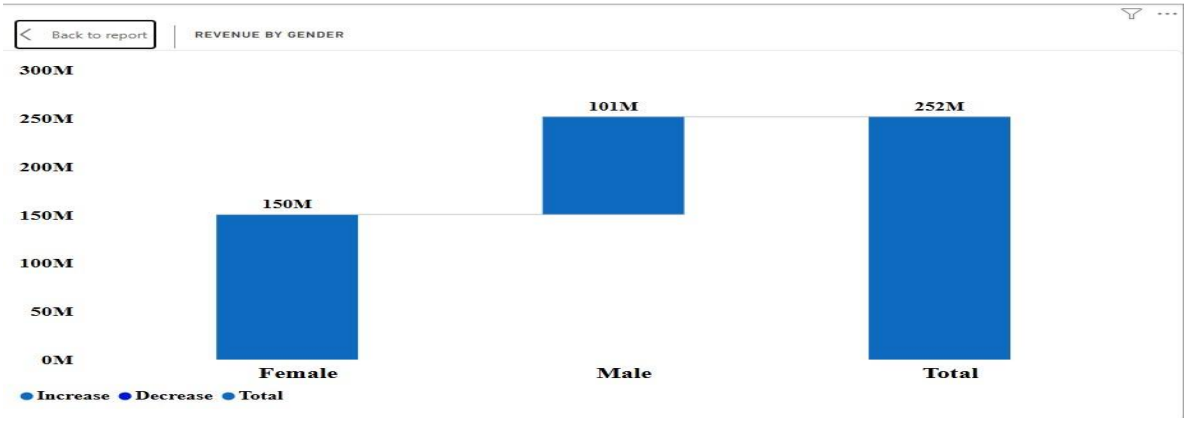
Categories include:

- Clothing
- Shoes
- Technology

- Cosmetics
- Toys
- Food & Beverage
- Books
- Souvenir

17.5 REVENUE BY GENDER:

Chart Type: Waterfall Chart



This visualization compares revenue generated by male and female customers.

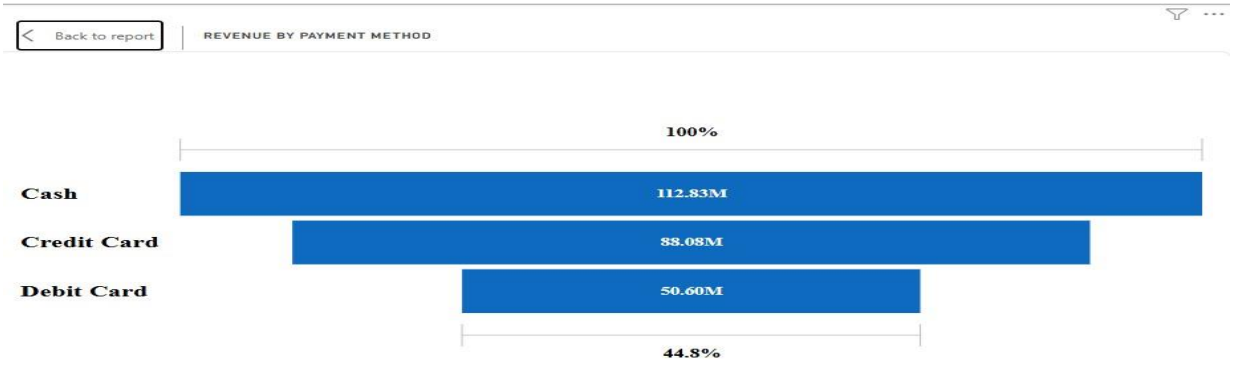
From the dataset:

- Female customer revenue ≈ 150.21M
- Male customer revenue ≈ 101.30M

This analysis helps understand revenue distribution across the two gender groups represented in the dataset.

17.6 REVENUE BY PAYMENT METHOD:

Chart Type: Funnel Chart



The dashboard compares revenue generated through:

- Cash
- Credit Card
- Debit Card

Revenue from the dataset is approximately:

Payment Method	Revenue
Cash	112.83M
Credit Card	88.08M
Debit Card	50.60M

This analysis helps identify the contribution of different payment methods to overall sales.

18. KEY ANALYSIS POINTS / INSIGHTS

Insight 1 – Overall Revenue:

The dataset generated approximately 251.51 million in total revenue.

Insight 2 – Product Categories:

Clothing generated the highest revenue among the eight product categories, followed by Shoes and Technology.

Insight 3 – Shopping Malls:

Mall of Istanbul and Kanyon generated the highest revenue among the shopping malls in the dataset.

Insight 4 – Gender:

Female customers contributed approximately 150.21M in revenue, while male customers contributed approximately 101.30M.

Insight 5 – Payment Method:

Cash transactions contributed approximately 112.83M, followed by Credit Card and Debit Card transactions.

Insight 6 – Quantity:

A total of approximately 298,712 products were purchased across the transactions.

Insight 7 – Average Order Value:

The average revenue per order is approximately 2.53K.

19. DASHBOARD FEATURES

The dashboard provides a simple visual representation of the data.

Features include:

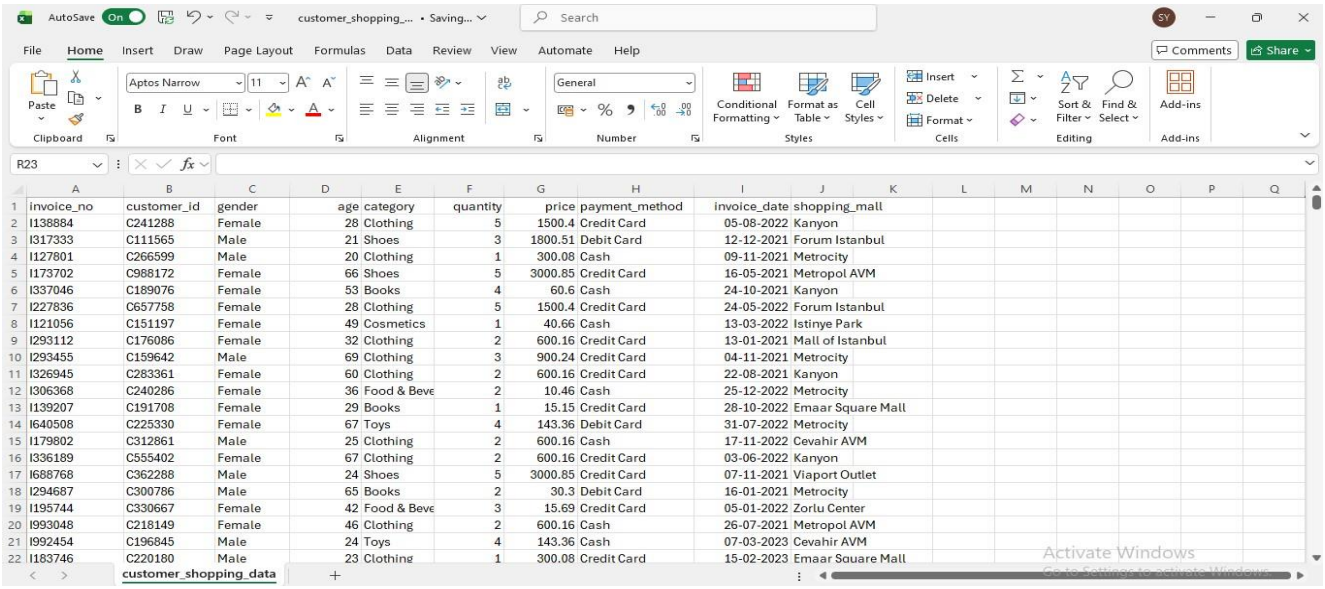
- KPI cards
- Category-wise analysis
- Shopping mall analysis
- Gender analysis
- Payment method analysis
- Customer category analysis
- Time-based analysis
- Interactive Power BI visuals
- Easy comparison of sales performance

20. TOOLS AND TECHNOLOGIES USED

Tool	Purpose
Microsoft Excel/CSV	Dataset
Power Query	Data cleaning and transformation
Microsoft Power BI	Data analysis and visualization
DAX	Measures and calculations

21. OUTPUT SCREENS

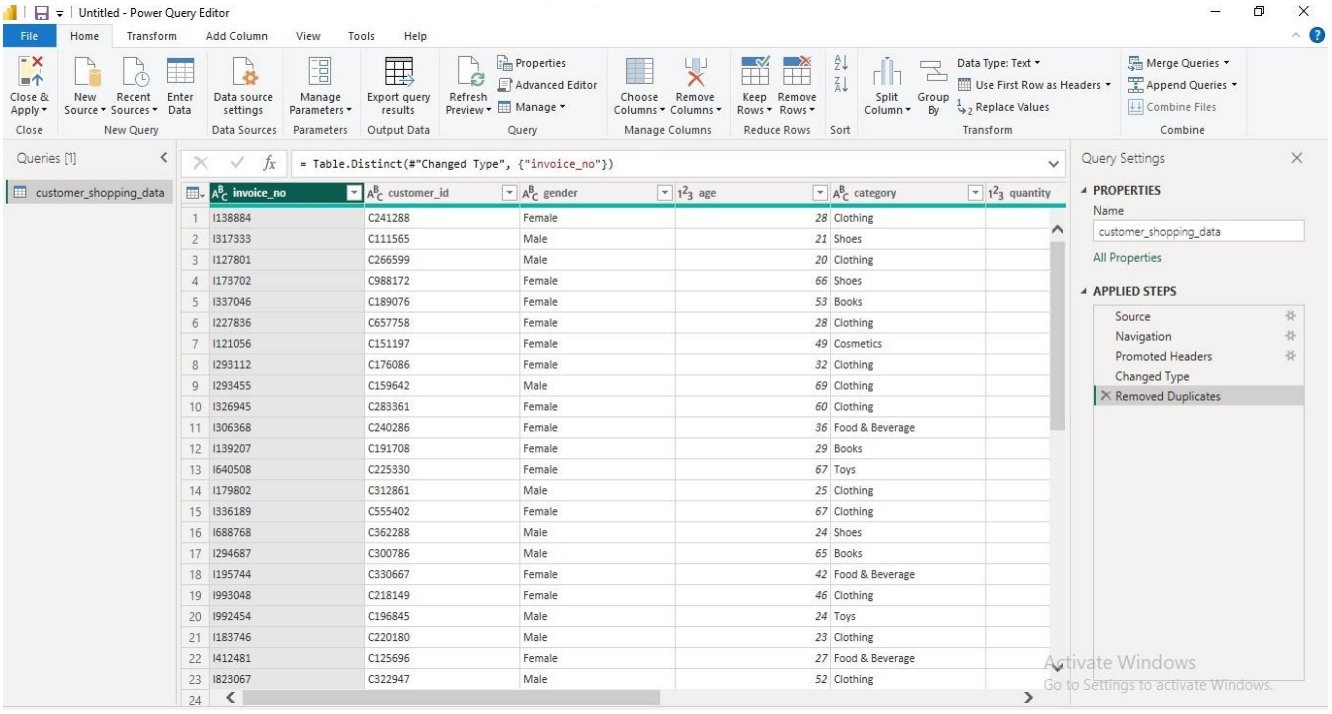
1. Raw Dataset:



	A	B	C	D	E	F	G	H	I	J
1	invoice_no	customer_id	gender	age	category	quantity	price	payment_method	invoice_date	shopping_mall
2	I138884	C241288	Female	28	Clothing	5	1500.4	Credit Card	05-08-2022	Kanyon
3	I317333	C111565	Male	21	Shoes	3	1800.51	Debit Card	12-12-2021	Forum Istanbul
4	I127801	C266599	Male	20	Clothing	1	300.08	Cash	09-11-2021	Metrocity
5	I173702	C988172	Female	66	Shoes	5	3000.85	Credit Card	16-05-2021	Metropol AVM
6	I337046	C189076	Female	53	Books	4	60.6	Cash	24-10-2021	Kanyon
7	I227836	C657758	Female	28	Clothing	5	1500.4	Credit Card	24-05-2022	Forum Istanbul
8	I121056	C151197	Female	49	Cosmetics	1	40.66	Cash	13-03-2022	Istinye Park
9	I293112	C176086	Female	32	Clothing	2	600.16	Credit Card	13-01-2021	Mall of Istanbul
10	I293455	C159642	Male	69	Clothing	3	900.24	Credit Card	04-11-2021	Metrocity
11	I326945	C283361	Female	60	Clothing	2	600.16	Credit Card	22-08-2021	Kanyon
12	I306368	C240286	Female	36	Food & Beve	2	10.46	Cash	25-12-2022	Metrocity
13	I139207	C191708	Female	29	Books	1	15.15	Credit Card	28-10-2022	Emaar Square Mall
14	I640508	C225330	Female	67	Toys	4	143.36	Debit Card	31-07-2022	Metrocity
15	I179802	C312861	Male	25	Clothing	2	600.16	Cash	17-11-2022	Cevahir AVM
16	I336189	C555402	Female	67	Clothing	2	600.16	Credit Card	03-06-2022	Kanyon
17	I688768	C362288	Male	24	Shoes	5	3000.85	Credit Card	07-11-2021	Viaport Outlet
18	I294687	C300786	Male	65	Books	2	30.3	Debit Card	16-01-2021	Metrocity
19	I195744	C330667	Female	42	Food & Beve	3	15.69	Credit Card	05-01-2022	Zorlu Center
20	I993048	C218149	Female	46	Clothing	2	600.16	Cash	26-07-2021	Metropol AVM
21	I992454	C196845	Male	24	Toys	4	143.36	Cash	07-03-2023	Cevahir AVM
22	I183746	C220180	Male	23	Clothing	1	300.08	Credit Card	15-02-2023	Emaar Square Mall

Figure 1: Raw Customer Shopping Dataset

2. Power Query:



	invoice_no	customer_id	gender	age	category	quantity
1	I138884	C241288	Female	28	Clothing	
2	I317333	C111565	Male	21	Shoes	
3	I127801	C266599	Male	20	Clothing	
4	I173702	C988172	Female	66	Shoes	
5	I337046	C189076	Female	53	Books	
6	I227836	C657758	Female	28	Clothing	
7	I121056	C151197	Female	49	Cosmetics	
8	I293112	C176086	Female	32	Clothing	
9	I293455	C159642	Male	69	Clothing	
10	I326945	C283361	Female	60	Clothing	
11	I306368	C240286	Female	36	Food & Beverage	
12	I139207	C191708	Female	29	Books	
13	I640508	C225330	Female	67	Toys	
14	I179802	C312861	Male	25	Clothing	
15	I336189	C555402	Female	67	Clothing	
16	I688768	C362288	Male	24	Shoes	
17	I294687	C300786	Male	65	Books	
18	I195744	C330667	Female	42	Food & Beverage	
19	I993048	C218149	Female	46	Clothing	
20	I992454	C196845	Male	24	Toys	
21	I183746	C220180	Male	23	Clothing	
22	I412481	C125696	Female	27	Food & Beverage	
23	I823067	C322947	Male	52	Clothing	

Figure 2: Data Cleaning and Transformation in Power Query

3. Data Model:

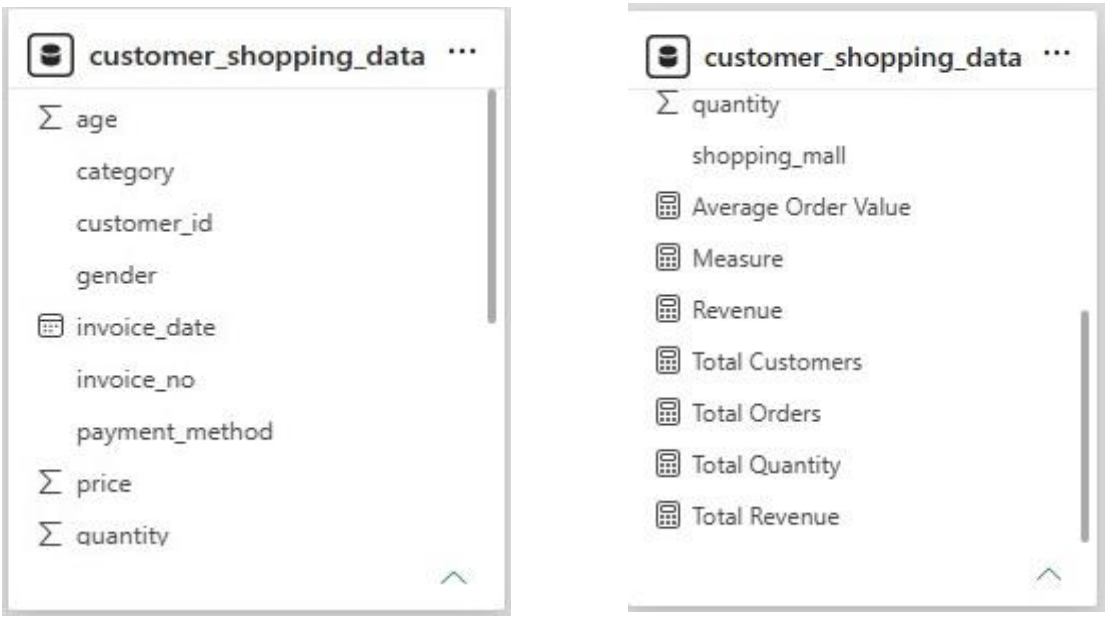


Figure 3: Power BI Model View

4. Dashboard:



Figure 4: Customer Sales Report Power BI Dashboard

22. CONCLUSION

The Customer Sales Report project demonstrates how raw customer transaction data can be transformed into meaningful business information using Microsoft Power BI.
The dataset was examined, cleaned, transformed, and analyzed before creating the dashboard. Various KPIs and visualizations were developed to understand revenue, customers, orders, quantity, categories, shopping malls, gender, payment methods, and time-based sales performance.

The interactive dashboard provides a clear and visual representation of the dataset and makes it easier to compare different business dimensions.

Overall, the project demonstrates the use of Power Query, data transformation, DAX calculations, data analysis, and Power BI visualization to convert raw data into an interactive business intelligence report.